Timeformer: Explicit Temporal Conditioning for Traceable Semantic Representations

Anonymous Author

Institution withheld for review

Abstract. Words change meaning over time, yet most computational models treat token representations as epoch-independent. We study temporal traceability: the property that the same surface token, queried at different epochs, occupies semantically appropriate neighborhoods in representation space. A model may use epoch information to improve predictive accuracy without exposing a traceable `token@time` representation: this gap is the focus of the paper.
We address it with Timeformer, a family of Transformer-style encoders organized as a controlled ablation chain. Four variants add temporal capacity in successive steps: a static baseline, a globally time-conditioned encoder, a token-time encoder that conditions each token embedding on the epoch before self-attention is applied, and a memory-augmented extension that consults causal historical prototypes. All variants are trained with masked language modeling on a controlled synthetic benchmark in which the temporal trajectory of each target item is known and evaluation labels are withheld from training.
The main finding is that explicit token-time interaction is the effective mechanism for traceability. The Token-Time Transformer produces a neighborhood drift of -0.614 for drifting subjects across epochs, compared to -0.231 for the static baseline (a difference stable across seeds and visible in class-specific trajectory curves). Probe-based accuracy gains are modest and seed-sensitive, confirming that geometric diagnostics are more appropriate than label recoverability for evaluating traceability. The memory-augmented variant does not produce a robust improvement over token-time interaction; the oracle diagnostic identifies mean-prototype aggregation as the failure point, motivating sense-aware memory mechanisms in future work.

Keywords: Temporal semantics · Semantic change · Transformer models · Traceability · Diagnostic evaluation

1 Introduction

Words do not preserve fixed meanings across time. In 1900, *broadcast* referred to scattering seeds across a field; by mid-century it denoted radio and television transmission; today it describes a social media post reaching millions of users. Most language models, however, treat word meanings as fixed: the same word produces the same embedding regardless of when it was used. The problem this creates is not merely that models ignore historical context. It is that the same word at different times does not carry the same meaning, and a model that cannot distinguish `broadcast@1900` from `broadcast@2020` cannot be inspected for semantic movement across time.

The ability to query a word at a specific epoch matters beyond technical benchmarks. Consider *gay*, which in the early twentieth century was predominantly associated with terms such as *cheerful*, *merry*, and *carefree*, and by the late twentieth century with *queer*, *lesbian*, and *bisexual* [3]. Whether this transition involved a shift toward or away from terms linked to social stigma, and when exactly that shift occurred, is an empirical question. To answer it, one needs a representation that exposes `gay@1920` and `gay@1980` as directly comparable objects in a shared space. Without such a representation, the question can only be addressed through aggregate statistics that conflate meaning with unrelated features of the corpus.

The dominant computational approach, lexical semantic change detection (LSCD), measures this kind of shift by training distributional representations on time-sliced corpora and comparing them across periods [6, 7]. One model may associate *broadcast* in 1900 with *sow* and *field*, while another, trained on 2020 data, associates it with *signal* and *audience*; semantic change is inferred from the distance between the two spaces. This approach has produced important methods and benchmarks, but it summarizes the change as a single similarity score between two separately trained models, not as a queryable `token@time` object. `broadcast@1900` and `broadcast@2020` are implicit in the models but not directly inspectable within a single shared space.

A subtler limitation arises when a single model receives temporal information during training. A model may correctly interpret *virus* as biological infection in a 1900 sentence and as malicious software in a 2020 sentence, improving its word-prediction accuracy, while still assigning nearly identical neighborhoods to the token across epochs. Temporal information can improve prediction without producing a representation in which the same word occupies measurably different neighborhoods at different epochs. Prediction accuracy and geometric traceability are separable properties.

We call a representation *temporally traceable* if the same surface token, queried at different epochs, occupies neighborhoods consistent with its meaning at that epoch, making `token@time` a directly inspectable object. This paper introduces Timeformer, a family of Transformer-style encoders designed around this property. The family is organized as a controlled ablation chain: a Static Transformer with no time information; an Additive Time-Conditioned Transformer that injects a global epoch vector; a Token-Time Transformer that conditions each token embedding on the epoch before self-attention is applied; and a Memory-Augmented Timeformer that consults causal historical prototypes. This design lets us isolate which form of temporal conditioning produces traceability, rather than treating it as a single architectural choice. We study this in a controlled synthetic benchmark in which the temporal trajectory of each subject is known, giving diagnostic access to target neighborhoods without the confounds of natural historical corpora.

The paper makes four contributions: it formulates temporal traceability as a `token@time` representation problem; introduces the Timeformer ablation family; evaluates traceability with diagnostics that inspect subject representations directly (context drift score, contrastive sign-flip rate, and probe-based evaluation); and shows that token-time interaction is the main positive mechanism, while mean-pooled historical memory is better understood as a diagnostic extension. Section 2 reviews related work; Sections 3–5 describe the benchmark, architecture, and evaluation protocol; Section 6 reports results; Sections 7–8 discuss and conclude.

2 Related Work

Computational approaches to semantic change are commonly framed as lexical semantic change detection: given language data from different periods, the goal is to determine whether and how the meaning of a target item has changed. Diachronic word embeddings provided one of the most influential formulations of this problem by learning distributional representations for different time slices and comparing them across periods [3]. This family of methods made semantic change operational through geometric movement in embedding space, often after alignment or normalization across independently trained temporal slices. The central intuition remains important for this paper: semantic change should be visible as a change in neighborhood structure. Our focus, however, is not on aligning independently trained spaces. Instead, we ask whether a single Transformer-style encoder can expose a representation that is directly queryable as a lexical item at a specific epoch.

Benchmarks for LSCD have helped standardize evaluation by defining target words, periods, and gold annotations for semantic change [6, 7]. These resources are essential for evaluating change in natural corpora, but they inherit the observational complexity of historical language, where changes in frequency, genre, and corpus composition interact with lexical meaning. We use a controlled benchmark as a diagnostic testbed rather than a replacement for natural LSCD benchmarks, trading observational realism for direct access to planted trajectories and evaluation labels.

A second relevant line of work concerns temporal conditioning in neural models. Rather than training separate models for each period, a temporally conditioned model can receive time as part of its input or architecture, allowing parameter sharing across epochs while representations or predictions vary with time [1, 5, 4]. For the traceability problem, however, the form of conditioning matters. A global epoch vector may tell the model which period it is processing, but it does not necessarily force an interaction between a particular token identity and that epoch. Timeformer is therefore organized as an ablation chain that separates global time injection from explicit token-time interaction. This distinction is important because the paper evaluates not only whether time improves a prediction, but whether the same token occupies different and interpretable neighborhoods at different epochs.

Finally, this paper draws on diagnostic evaluation and multi-sense representations. Probe accuracy does not establish traceability, so we combine it with geometric diagnostics that inspect the subject token directly. Semantic change may involve bifurcation rather than a simple shift [2]; the memory-augmented variant exposes why mean-pooled prototypes are insufficient for that structure.

3 Controlled Benchmark Summary

We evaluate Timeformer on a controlled synthetic benchmark that makes temporal semantic movement observable. The benchmark provides a diagnostic environment with known target trajectories and representation-level comparison against planted ground truth; full construction details are in the companion benchmark paper.

The corpus uses a strict subject–verb–object (SVO) grammar. This structure isolates subjects as the lexical items with planted temporal trajectories, while verbs and objects serve as noisy contextual evidence. A subject's meaning at a given epoch is operationalized as the probability that its sentence is generated from one of two latent semantic contexts: $P(A \mid s, t)$ denotes the context-A probability for subject s at epoch t. A subject undergoes semantic change when $P(A \mid s, t)$ shifts across epochs. The two contexts are defined by disjoint sets of verbs and objects. The vocabulary contains 30 subjects, 8 verbs, and 8 objects; context A contains verbs $\{V1, \ldots, V4\}$ and objects $\{O1, \ldots, O4\}$, while context B contains verbs $\{V5, \ldots, V8\}$ and objects $\{O5, \ldots, O8\}$.

Subjects are divided into three planted trajectory classes. Stable subjects maintain an approximately constant $P(A \mid s, t)$ across all epochs. Drift subjects follow a monotonic trajectory from near-exclusive context A ($P(A \mid s, t) \approx 1$ at $t0$) toward near-exclusive context B ($P(A \mid s, t) \approx 0$ at $t9$). Bifurcating subjects begin in a dominant context A state and move toward a mixed regime ($P(A \mid s, t) \approx 0.5$) in which both contexts coexist with roughly equal weight. Full trajectory specifications are given in the companion benchmark paper; here, these classes define the expected direction and magnitude of neighborhood movement used in evaluation. The benchmark contains 10 epochs, $t0$ through $t9$, and 10 subjects per trajectory class. This design lets us test whether a model distinguishes stability, directional drift, and movement toward a mixed semantic state.

The observed verb and object are informative but deliberately noisy. For the main corpus, if the planted `true_context` is A, each local marker is sampled from context A with probability $p_{canon} = 0.75$ and from context B with probability 0.25; the reverse holds when the planted context is B. Thus, local evidence usually points in the intended direction but cannot be used as a deterministic oracle. The `true_context` label is defined by the planted subject–epoch trajectory, not by the observed verb or object, and is used only for evaluation.

Table 1. Model variants in the Timeformer ablation family.

ID	Public name	Temporal signal	Memory
B1	Static Transformer	None	No
B2a	Additive Time-Conditioned Transformer	Global epoch vector	No
B2b	Token-Time Transformer	Token-time interaction	No
B3	Memory-Augmented Timeformer	Token-time interaction	Causal prototypes

The benchmark contains several evaluation conditions. The standard test split uses the same noisy marker process as training. Harder splits include cases in which the verb contradicts the planted context and cases in which both local markers contradict it. An additional ambiguous split is generated with $p_{canon} = 0.50$, making verb and object markers locally uninformative; in this setting, the subject identity and epoch are the relevant evidence. We also use contrastive evaluations, in which the same subject is presented under alternative epoch labels to test whether the representation inverts accordingly, and continuation evaluations, in which the model is tested on epochs held out from the main training distribution. Both are described in detail in Section 5.

The benchmark plays two roles: it gives direct access to target trajectories, allowing us to ask whether a subject's representation at $t0$ and $t9$ occupies the intended semantic neighborhoods; and it separates context-local cues from temporal cues, distinguishing models that exploit observed verbs and objects from models that encode the interaction between lexical identity and epoch. Results are diagnostic evidence in a controlled setting, not a claim of performance on open-domain historical language.

4 Timeformer Architecture

Timeformer is defined as an ablation family rather than as a single model. All variants share the same basic training interface: an input sentence is encoded as `[CLS] S V O [SEP]`, a Transformer encoder produces contextual hidden states, and an MLM head predicts masked tokens. The representation used for traceability diagnostics is h_s, the final hidden state at the subject position. The variants differ only in how epoch information is made available to the encoder and whether the current subject representation can consult causal historical prototypes. Table 1 summarizes the four variants.

4.1 Static Transformer

The Static Transformer is the baseline encoder. It receives token and position embeddings but no explicit epoch information:

$$x_i = e(w_i) + p_i.$$

The resulting sequence is passed through a small pre-layer-normalized Transformer encoder. This model can learn corpus-level co-occurrence patterns, but it has no direct way to distinguish the representation of a subject at $t0$ from the same subject at $t9$, except through the local words observed in the sentence. It therefore represents the null hypothesis that temporal behavior can be recovered from static distributional evidence alone.

4.2 Continuous Time Encoding

The temporally conditioned variants use a continuous encoding of epoch index rather than a learned lookup table. For an epoch $t \in \{0, \ldots, T\}$, the index is normalized to $[0, 1]$, mapped to fixed sinusoidal features, and then projected by a small MLP:

$$\tau(t) = \mathrm{MLP}(\mathrm{sinusoidal}(t/T)).$$

This supports interpolation and extrapolation: a learned embedding table would memorize training epochs as unrelated identifiers, whereas a continuous encoding treats them as ordered points on a shared scale. These sinusoidal features encode epoch index only and are distinct from the token-position encodings of the Transformer.

4.3 Additive Time-Conditioned Transformer

The Additive Time-Conditioned Transformer injects the same epoch vector into every token position:

$$x_i = e(w_i) + p_i + \tau(t).$$

This model tests the minimal effect of exposing the encoder to the current epoch. Because the same temporal offset is added to every token, the model is given global period information but no token-specific temporal modulation at the embedding layer. It therefore isolates the question of whether simply telling the model the epoch is sufficient for traceability.

4.4 Token-Time Transformer

The Token-Time Transformer replaces additive conditioning with an explicit interaction between lexical identity and epoch. For each token, the token embedding and time encoding are concatenated and projected back to the model dimension:

$$x_i = W[e(w_i); \tau(t)] + p_i.$$

Here $e(w_i), \tau(t) \in \mathbb{R}^d$, so $W \in \mathbb{R}^{d \times 2d}$ projects the concatenation back to the model dimension. Unlike the additive variant, this allows the same epoch vector to affect different tokens differently before self-attention: the same subject receives a distinct embedding at each epoch, which is the mechanism the paper evaluates for traceability.

Figure 2. Timeformer architecture detail

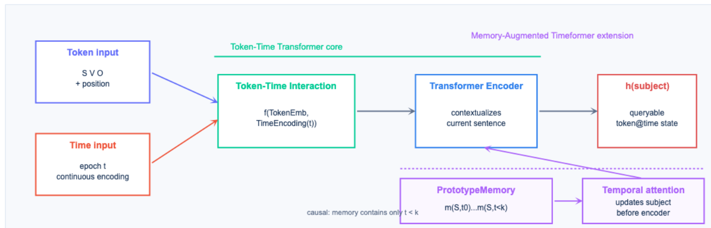

Fig. 1. Architecture detail for the Token-Time Transformer and the optional memory-augmented extension. The memory module attends only to causal historical prototypes and updates the subject representation before Transformer encoding.

4.5 Memory-Augmented Timeformer

The Memory-Augmented Timeformer extends the Token-Time Transformer with a causal prototype memory. For each subject s and epoch t, a prototype $m(s,t)$ is computed as the mean of subject representations observed in training sentences for that subject and epoch. Prototypes are computed with stop-gradient, from training data only, and are served causally: at epoch t_k, the model may attend only to $\{m(s,t) : t < t_k\}$.

The memory module is a temporal cross-attention layer applied only to the subject embedding. The current subject embedding acts as the query, and the historical prototypes for the same subject act as keys and values. The updated subject embedding is then inserted back into the token sequence before the Transformer encoder:

$$\tilde{x}_s = x_s + g \cdot (\mathrm{LN}(x_s + \mathrm{Attn}(x_s, M_s, M_s)) - x_s)\,,$$

where M_s is the causal history for subject s, and $g \in [0,1]$ is a learned scalar gate initialized to zero so that at the start of training $\tilde{x}_s = x_s$ and the memory path is inactive. The formula interpolates between the original embedding ($g = 0$) and the layer-normalized attended representation ($g = 1$), allowing the model to gradually open the memory channel as training progresses. Applying the memory update before the textual encoder lets the subsequent self-attention layers propagate historical information from the subject to the other token positions, creating a path by which the MLM objective can reward useful memory use.

5 Evaluation Protocol

All variants are trained with masked language modeling over the synthetic SVO corpus. The planted `true_context` label is never used during training; it is used only after training to evaluate whether the learned representation contains temporal semantic information. This separation reflects the intended use case of representation learning from corpora without semantic-change labels. The primary evaluation target is h_s, the final hidden state at the subject position, since subjects are the lexical items with planted temporal trajectories.

We use four complementary diagnostics. First, a linear probe is trained on h_s from the standard test split to predict `true_context` and is then evaluated on each split. This measures whether context information is linearly recoverable from the representation, but it does not by itself establish traceability. Second, we compute a context drift score over nearest neighbors. For each epoch and subject class, we retrieve the k nearest neighbors of each subject representation and measure the proportion assigned to context A, with k fixed at the same value across all models and splits. A traceable representation should show a strong movement from A-like to B-like neighborhoods for drifting subjects, weaker movement for bifurcating subjects, and comparatively limited movement for stable subjects. Third, we use contrastive temporal examples in which the same subject token is presented with two different epoch labels. We define the contrastive sign-flip rate as the proportion of such pairs for which swapping the epoch label reverses the nearest-neighbor context assignment (from A-dominant to B-dominant or vice versa). A model without temporal capacity is expected to produce a sign-flip rate of zero, since the same token yields the same representation regardless of the epoch label provided. Fourth, we evaluate continuation examples from held-out later epochs to test whether temporal behavior extends beyond the training distribution.

The ablation comparisons follow the architecture chain. The gain from time-conditioning is measured by comparing the Additive Time-Conditioned Transformer to the Static Transformer, primarily on the ambiguous split where local markers are uninformative. The gain from token-time interaction is measured by comparing the Token-Time Transformer to the additive variant on the same split. The memory contribution is measured by comparing the Memory-Augmented Timeformer to the Token-Time Transformer on continuation and contrastive diagnostics.

For the memory-augmented model, we include a shuffled-memory control (a subject receives prototypes from a different subject), a no-history control (the memory path is present but empty), and an oracle-memory diagnostic (the model receives perfect historical prototypes). Memory is considered useful only when it improves over the Token-Time Transformer and over both inappropriate and empty histories.

Fig. 2. Context drift score across epochs. The Token-Time Transformer shows stronger movement from context A toward context B for drifting subjects than the Static Transformer, while stable and bifurcating subjects provide class-specific controls.

6 Results

We report results over three training runs. Unless otherwise noted, values are means over seeds, with the standard deviation shown in parentheses when space allows. The main finding is that token-time interaction produces a much clearer temporal movement in representation space than the Static Transformer. The memory-augmented variant is informative as a diagnostic extension, but it does not provide a robust improvement over the Token-Time Transformer.

6.1 Temporal Neighborhood Movement

Figure 2 and Table 2 show the context drift score across epochs. For drifting subjects, the Static Transformer drops from 0.618 to 0.387 ($\Delta = -0.231$), while the Token-Time Transformer shows a much stronger transition from 0.847 to 0.233 ($\Delta = -0.614$). This is the clearest evidence for temporal traceability: the same subject token occupies substantially different neighborhoods when queried at different epochs.

The class-specific curves serve as controls (Table 2). Stable subjects show much smaller movement and no coherent A-to-B trend. Bifurcating subjects show intermediate drift consistent with a trajectory toward a mixed state, not a complete transition. The geometric diagnostic thereby distinguishes all three trajectory classes without relying on a single aggregate score.

6.2 Probe and Contrastive Diagnostics

The linear probe results are supportive but less decisive than the neighborhood analysis. On the ambiguous split, where local verb and object markers are uninformative, the Static Transformer reaches 0.577 accuracy, the Additive Time-Conditioned Transformer reaches 0.573, and the Token-Time Transformer reaches 0.599. The corresponding mean delta for token-time interaction is

Table 2. Context drift score at the first and last epochs. Values are context-A proportions among nearest neighbors; $\Delta = t9 - t0$.

Class	Model	t0	t9	Δ
Stable	Static Transformer	0.736	0.756	+0.020
Stable	Token-Time Transformer	0.843	0.641	-0.203
Stable	Memory-Augmented Timeformer	0.810	0.658	-0.153
Drift	Static Transformer	0.618	0.387	-0.231
Drift	Token-Time Transformer	0.847	0.233	-0.614
Drift	Memory-Augmented Timeformer	0.836	0.263	-0.573
Bifurcation	Static Transformer	0.794	0.682	-0.112
Bifurcation	Token-Time Transformer	0.895	0.521	-0.374
Bifurcation	Memory-Augmented Timeformer	0.905	0.517	-0.388

Table 3. Main traceability diagnostics. Accuracy columns report linear-probe accuracy on h_s; flip is contrastive sign-flip rate. A dash indicates the metric was not computed for this variant.

Model	Drift	Flip	Ambig.	Cont.
Static Transformer	-0.231	0.000	0.577	0.719
Additive Time-Conditioned Transformer	–	0.690	0.573	0.754
Token-Time Transformer	-0.614	0.517	0.599	0.766
Memory-Augmented Timeformer	-0.573	0.506	0.599	0.766

+0.026, but it is seed-sensitive: the three runs give +0.053, +0.005, and +0.021. This confirms that probe accuracy alone should not carry the central claim. It measures recoverability of the planted label, not the geometry of temporal movement.

The contrastive sign-flip diagnostic gives a different view. The Static Transformer has a sign-flip rate of 0.000, as expected. The Additive variant averages 0.690 while the Token-Time Transformer averages 0.517, an ordering that may appear counterintuitive. The decoupling arises because sign-flip counts threshold crossings in individual contrastive pairs, whereas drift score measures the full trajectory magnitude. A global additive offset displaces all tokens uniformly, generating many crossings without coherent subject-level movement. Token-time interaction produces more structured, subject-specific movement (evidenced by the stronger drift) but does not necessarily cross the threshold in more pairs. We therefore treat sign flip as evidence that time conditioning changes behavior at all, and the neighborhood trajectories as the more diagnostic measure of traceability.

6.3 Memory Diagnostics

The Memory-Augmented Timeformer does not produce a robust gain over the Token-Time Transformer. On continuation, both models average 0.766, and the per-seed memory deltas are +0.013, +0.005, and −0.018. The oracle diagnostic clarifies the failure mode: oracle memory improves contrastive sign flip (0.586

Table 4. Memory diagnostics from the corrected oracle run. Continuation is linear-probe accuracy on held-out later epochs; flip is contrastive sign-flip rate.

Variant	Continuation	Sign flip	Delta vs. B2b ref.
B2b reference	0.762	0.448	–
B3 learned	0.775	0.448	+0.013 cont.; +0.000 flip
B3 oracle	0.754	0.586	-0.008 cont.; +0.138 flip

vs. 0.448 for the Token-Time reference) but not continuation (0.754 vs. 0.762). Perfect historical prototypes can affect contrastive temporal behavior, but do not solve the continuation objective under the current MLM setup, supporting the use of B3 as a diagnostic extension.

7 Discussion

The results support a clear conclusion: temporal traceability is a geometric property that does not reduce to predictive accuracy. The Token-Time Transformer improves probe accuracy by only +0.026 on average (modest and seed-sensitive), yet shows a neighborhood drift of −0.614 vs. −0.231 for the static baseline. This dissociation is the central empirical finding: a model can use epoch information to improve predictions without exposing a representation in which the same lexical item occupies meaningfully different neighborhoods at different epochs. Evaluating traceability therefore requires geometric diagnostics, not only label recoverability.

The additive variant shows that knowing the epoch is not sufficient. A global additive offset shifts all token representations by the same vector, leaving subject-level trajectory geometry weakly constrained. Token-time interaction changes this: concatenating each token embedding with the epoch encoding before projection allows the same epoch to affect different tokens differently, so the same subject receives a distinct embedding at each epoch. This is the mechanism behind the coherent neighborhood trajectories in the drift analysis.

The Memory-Augmented Timeformer tests the simplest form of causal memory: a mean of historical subject representations per epoch. Learned prototypes produce near-zero average gain over the Token-Time Transformer, and the oracle diagnostic identifies why: perfect historical prototypes improve contrastive sign-flip rate but not continuation accuracy. For bifurcating subjects, which develop two coexisting senses in later epochs, the mean prototype is a centroid between two clusters, representing neither sense well. This motivates sense-aware memory mechanisms, studied in the companion paper, that preserve the internal structure of historical representations rather than collapsing it.

8 Conclusion

This paper studied whether a Transformer-style encoder can expose a temporally traceable representation, meaning one in which the same lexical item, queried at different epochs, occupies neighborhoods that reflect its intended temporal context. The answer, in the controlled benchmark used here, is conditional: explicit token-time interaction is necessary. A static encoder and a globally time-conditioned encoder both fail to produce coherent subject-level trajectories. An encoder that conditions each token embedding on the epoch before self-attention is applied does produce such trajectories, with a neighborhood drift for drifting subjects that is more than twice that of the static baseline (-0.614 vs. -0.231).

The contribution is narrow and deliberate: token-time interaction supports traceability in a controlled diagnostic setting, not a claim about open-domain historical language. The evaluation protocol also contributes a method for separating representational traceability from predictive performance. Mean-pooled historical prototypes do not produce a robust improvement, and the oracle diagnostic identifies single-prototype aggregation as the failure point. Sense-aware memory mechanisms are the natural next step and are studied in the companion paper.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Dhingra, B., Cole, J.R., Eisenschlos, J.M., Gillick, D., Eisenstein, J., Cohen, W.W.: Time-aware language models as temporal knowledge bases. Transactions of the Association for Computational Linguistics (TACL) **10**, 257–273 (2022)
2. Giulianelli, M., Del Tredici, M., Fernández, R.: Analysing lexical semantic change with contextualized word representations. In: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL). pp. 3960–3973 (2020)
3. Hamilton, W.L., Leskovec, J., Jurafsky, D.: Diachronic word embeddings reveal statistical laws of semantic change. In: Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (ACL). pp. 1489–1501 (2016)
4. Loureiro, D., Barbieri, F., Neves, L., Anke, L.E., Camacho-Collados, J.: TimeLMs: Diachronic language models from Twitter. In: Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL): System Demonstrations. pp. 251–260 (2022)
5. Rosin, G., Radinsky, K.: Time masking for temporal language models. In: Proceedings of the 15th ACM International Conference on Web Search and Data Mining (WSDM). pp. 833–841 (2022)
6. Schlechtweg, D., McGillivray, B., Hengchen, S., Dubossarsky, H., Tahmasebi, N.: SemEval-2020 task 1: Unsupervised lexical semantic change detection. In: Proceedings of the 14th International Workshop on Semantic Evaluation (SemEval). pp. 1–23 (2020)
7. Tahmasebi, N., Borin, L., Jatowt, A.: Survey of computational approaches to lexical semantic change. Computational Approaches to Semantic Change pp. 1–91 (2021)